

Task 24:

Use Vibe Coding to build a complete cloud-based feature using Generative AI. Write iterative prompts to generate a REST API, cloud deployment steps, and security configurations. Refine the prompts based on the output quality, document prompt versions, and evaluate how prompt engineering improves code accuracy, scalability, and real-world usability

Aim

To use **Vibe Coding and Prompt Engineering** to build a complete **cloud-based feature** using Generative AI by generating:

- REST API
- Cloud deployment steps
- Security configurations
and refining prompts iteratively to improve **accuracy, scalability, and real-world usability**.

Algorithm

1. Define a real-world feature (e.g., Product API deployment).
2. Write an **initial prompt**.
3. Generate basic output (API code).
4. Evaluate output (missing features).
5. Refine prompt to include:
 - Deployment steps
 - Security
 - Scalability
6. Generate improved output.
7. Repeat refinement (iteration).
8. Compare outputs across versions.
9. Document improvements.
10. Analyze effectiveness of prompt engineering.

Implementation

Generate a REST API for Product Management.

Generate a Spring Boot REST API with CRUD operations, MySQL database, and proper layered architecture.

Generate a cloud-ready Spring Boot REST API with:

- CRUD operations
- MySQL configuration
- Docker support
- AWS deployment steps
- Basic security (JWT)

FROM openjdk:17

COPY target/app.jar app.jar

ENTRYPOINT ["java","-jar","/app.jar"]

@Configuration

@EnableWebSecurity

public class SecurityConfig {

 @Bean

 public SecurityFilterChain filterChain(HttpSecurity http) throws Exception {

 http.csrf().disable()

 .authorizeHttpRequests(auth -> auth

 .anyRequest().authenticated()

)

 .httpBasic();

 return http.build();

 }

}

Output:

Iterative prompt refinement improved:

- Code quality
- Security implementation
- Deployment readiness
- Real-world usability

Result

Successfully built a **cloud-ready feature using AI** Applied **iterative prompt engineering**
Improved code accuracy and scalability Generated deployment and security configurations

